VERSALITY CHARACTERIZED BY THE COTANGENT COMPLEX COMPARISON

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4234-4265 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4267-4357, the complete author proof (89 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_artin.json (pb=4066, stmt 4234-4265, proof 4267-4357).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00IN: sources/stacks-artin/pinned-original-source.tex, source lines 4357–4357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YI: sources/stacks-artin/pinned-original-source.tex, source lines 4068–4091. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06H0: sources/stacks-artin/pinned-original-source.tex, source lines 4357–4357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06IU: sources/stacks-artin/pinned-original-source.tex, source lines 4357–4357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YK: sources/stacks-artin/pinned-original-source.tex, source lines 4150–4153. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YL: sources/stacks-artin/pinned-original-source.tex, source lines 4161–4173. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YC: sources/stacks-artin/pinned-original-source.tex, source lines 3806–3838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$ satisfying (RS*). Let $U = \mathrm{Spec}(A)$ be an affine scheme of finite type over S which maps into an affine open $\mathrm{Spec}(\Lambda)$. Let x be an object of $\mathcal{X}$ over U. Let $\xi : E \to NL_{A/\Lambda}$ be a morphism of $D^-(A)$. Assume*

(i) *for every deformation situation $(x, A' \to A)$ we have: x lifts to $\mathrm{Spec}(A')$ if and only if $E \to NL_{A/\Lambda} \to NL_{A/A'}$ is zero, and*
(ii) *there is an isomorphism of functors $T_x(-) \to \mathrm{Ext}^0_A(E, -)$ such that $E \to NL_{A/\Lambda} \to \Omega^1_{A/\Lambda}$ corresponds to the canonical element (see Remark 07YC).*

Let $u_0 \in U$ be a finite type point with residue field $k = \kappa(u_0)$. Consider the following statements

(1) *x is versal at u_0, and*
(2) *$\xi : E \to NL_{A/\Lambda}$ induces a surjection $H^{-1}(E \otimes^{\mathbf{L}}_A k) \to H^{-1}(NL_{A/\Lambda} \otimes^{\mathbf{L}}_A k)$ and an injection $H^0(E \otimes^{\mathbf{L}}_A k) \to H^0(NL_{A/\Lambda} \otimes^{\mathbf{L}}_A k)$.*

Then we always have (2) $\Rightarrow$ (1) and we have (1) $\Rightarrow$ (2) if u_0 is a closed point.

Proof. Let $\mathfrak{p} = \mathrm{Ker}(A \to k)$ be the prime corresponding to u_0.

Assume that x versal at u_0 and that u_0 is a closed point of U. If $H^{-1}(\xi \otimes^{\mathbf{L}}_A k)$ is not surjective, then let $A' \to A$ be an extension with kernel I as in Lemma 07YL. Because u_0 is a closed point, we see that I is a finite A-module, hence that A' is a finite type Λ-algebra (this fails if u_0 is not closed). In particular A' is Noetherian. By property (c) for A' and (i) for ξ we see that x lifts to an object x' over A'. Let $\mathfrak{p}' \subset A'$ be kernel of the surjective map to k. By Artin-Rees (Algebra, Lemma 00IN) there exists an $n > 1$ such that $(\mathfrak{p}')^n \cap I = 0$. Then we see that

$$B' = A'/(\mathfrak{p}')^n \longrightarrow A/\mathfrak{p}^n = B$$

is a small, essential extension of local Artinian rings, see Formal Deformation Theory, Lemma 06H0. On the other hand, as x is versal at u_0 and as $x'|_{\mathrm{Spec}(B')}$ is a lift of $x|_{\mathrm{Spec}(B)}$, there exists an integer $m \geq n$ and a map $q : A/\mathfrak{p}^m \to B'$ such that the composition $A/\mathfrak{p}^m \to B' \to B$ is the quotient map. Since the maximal ideal of B' has nth power equal to zero, this q factors through B which contradicts the fact that $B' \to B$ is an essential surjection. This contradiction shows that $H^{-1}(\xi \otimes^{\mathbf{L}}_A k)$ is surjective.

Assume that x versal at u_0. By Lemma 07YK the map $H^0(\xi \otimes^{\mathbf{L}}_A k)$ is dual to the map $\mathrm{Ext}^0_A(NL_{A/\Lambda}, k) \to \mathrm{Ext}^0_A(E, k)$. Note that

$$\mathrm{Ext}^0_A(NL_{A/\Lambda}, k) = \mathrm{Der}_\Lambda(A, k) \quad \text{and} \quad T_x(k) = \mathrm{Ext}^0_A(E, k)$$

Condition (ii) assures us the map $\mathrm{Ext}^0_A(NL_{A/\Lambda}, k) \to \mathrm{Ext}^0_A(E, k)$ sends a tangent vector θ to U at u_0 to the corresponding infinitesimal deformation of x_0, see Remark 07YC. Hence if x is versal, then this map is surjective, see Formal Deformation

Theory, Lemma 06IU. Hence $H^0(\xi \otimes^{\mathbf{L}}_A k)$ is injective. This finishes the proof of (1) $\Rightarrow$ (2) in case u_0 is a closed point.

For the rest of the proof assume $H^{-1}(E \otimes^{\mathbf{L}}_A k) \to H^{-1}(NL_{A/\Lambda} \otimes^{\mathbf{L}}_A k)$ is surjective and $H^0(E \otimes^{\mathbf{L}}_A k) \to H^0(NL_{A/\Lambda} \otimes^{\mathbf{L}}_A k)$ injective. Set $R = A^\wedge_{\mathfrak{p}}$ and let η be the formal object over R associated to $x|_{\mathrm{Spec}(R)}$. The map $d\underline{\eta}$ on tangent spaces is surjective because it is identified with the dual of the injective map $H^0(E \otimes^{\mathbf{L}}_A k) \to H^0(NL_{A/\Lambda} \otimes^{\mathbf{L}}_A k)$ (see previous paragraph). According to Formal Deformation Theory, Lemma 06IU it suffices to prove the following: Let $C' \to C$ be a small extension of finite type Artinian local Λ-algebras with residue field k. Let $R \to C$ be a Λ-algebra map compatible with identifications of residue fields. Let $y = x|_{\mathrm{Spec}(C)}$ and let y' be a lift of y to C'. To show: we can lift the Λ-algebra map $R \to C$ to $R \to C'$.

Observe that it suffices to lift the Λ-algebra map $A \to C$. Let $I = \mathrm{Ker}(C' \to C)$. Note that I is a 1-dimensional k-vector space. The obstruction ob to lifting $A \to C$ is an element of $\mathrm{Ext}^1_A(NL_{A/\Lambda}, I)$, see Example 07YI. By Lemma 07YK and our assumption the map ξ induces an injection

$$\mathrm{Ext}^1_A(NL_{A/\Lambda}, I) \longrightarrow \mathrm{Ext}^1_A(E, I)$$

By the construction of ob and (i) the image of ob in $\mathrm{Ext}^1_A(E, I)$ is the obstruction to lifting x to $A \times_C C'$. By (RS*) the fact that y/C lifts to y'/C' implies that x lifts to $A \times_C C'$. Hence $ob = 0$ and we are done. $\square$